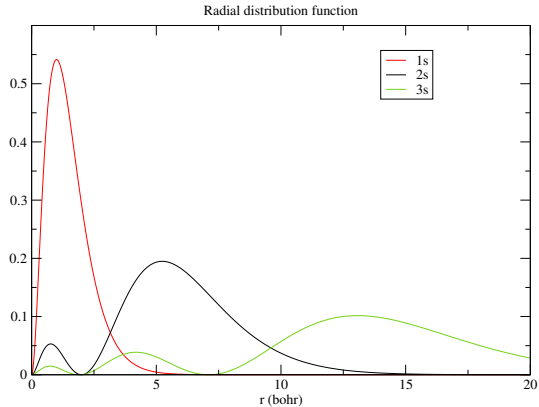


Strong local correlations and DFT+*U*

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CEA, DAM, DIF
Université Paris-Saclay, CEA, LMCE
Bruyères-le-Châtel, France

Abinit school 2026 - CEA

Atomic orbitals 1s, 2s and 3s



$$\Psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta, \phi)$$

Radial distribution function is $(rR_{nl}(r))^2$

4s/3d/4p

5s/4d/5p

Lanthanides 4fActinides 5f

4s/3d/4p	1	H																	2	He																
	3	Li	4	Be											5	B	6	C	7	N	8	O	9	F	10	Ne										
	11	Na	12	Mg											13	Al	14	Si	15	P	16	S	17	Cl	18	Ar										
	19	K	20	Ca	21	Sc	22	Ti	23	V	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn	31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
	37	Rb	38	Sr	39	Y	40	Zr	41	Nb	42	Mo	43	Tc	44	Ru	45	Rh	46	Pd	47	Ag	48	Cd	49	In	50	Sn	51	Sb	52	Te	53	I	54	Xe
	55	Cs	56	Ba	57-71	La-Lu	72	Hf	73	Ta	74	W	75	Re	76	Os	77	Ir	78	Pt	79	Au	80	Hg	81	Tl	82	Pb	83	Bi	84	Po	85	At	86	Rn
5s/4d/5p	87	Fr	88	Ra	89-103	Ac-Lr	104	Rf	105	Db	106	Sg	107	Bh	108	Hs	109	Mt	110	Ds	111	Rg	112	Cn	113	Nh	114	Fl	115	Mc	116	Lv	117	Ts	118	Og

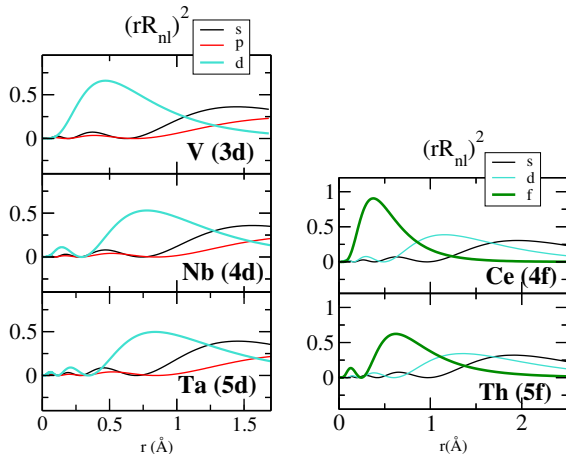
Lanthanides 4fActinides 5f

4s/3d/4p

5s/4d/5p

Lanthanides 4fActinides 5f

Localization of 3d, 4f and 5f orbitals.

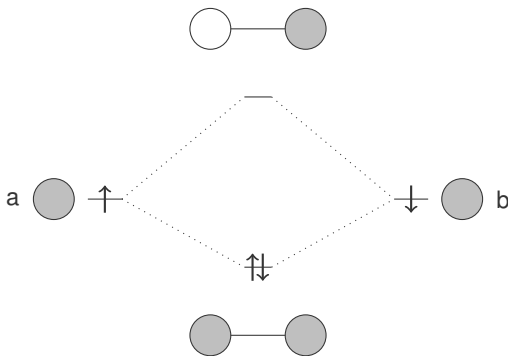


- $\Psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta, \phi)$
- For 1s, 2p, 3d, 4f, R has no node, their maxima are thus closer to the nucleus
- 3d and 4f orbitals are more localized.

Hydrogen Molecule

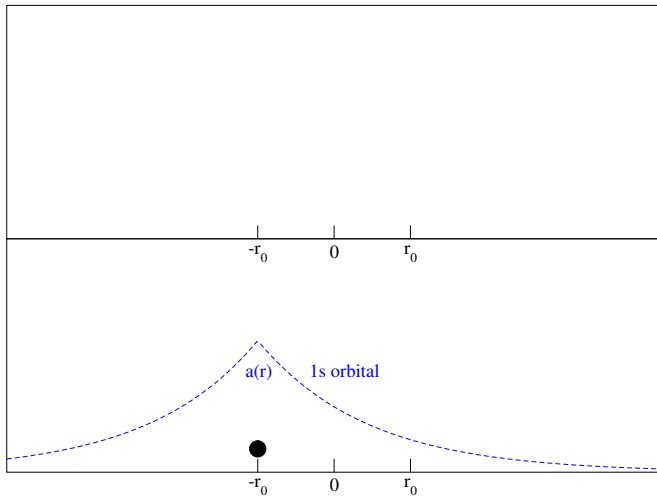
Non diagonal elements of the Hamiltonian creates chemical bonding

antibonding state $\sigma_u = \frac{a-b}{\sqrt{2}}$

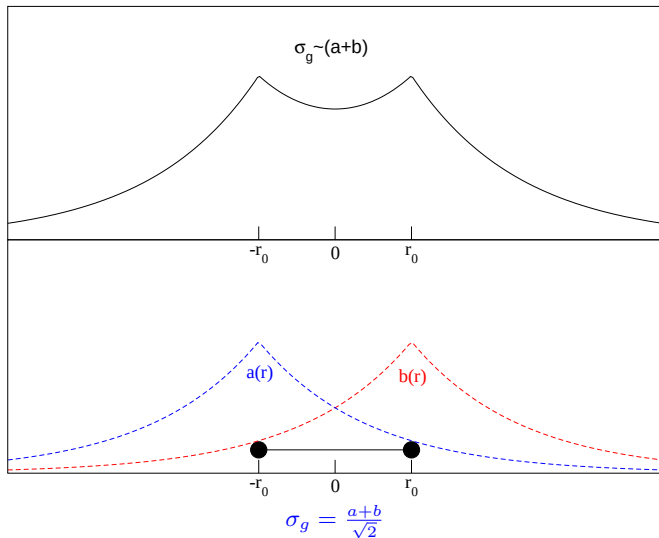


bonding state $\sigma_g = \frac{a+b}{\sqrt{2}}$

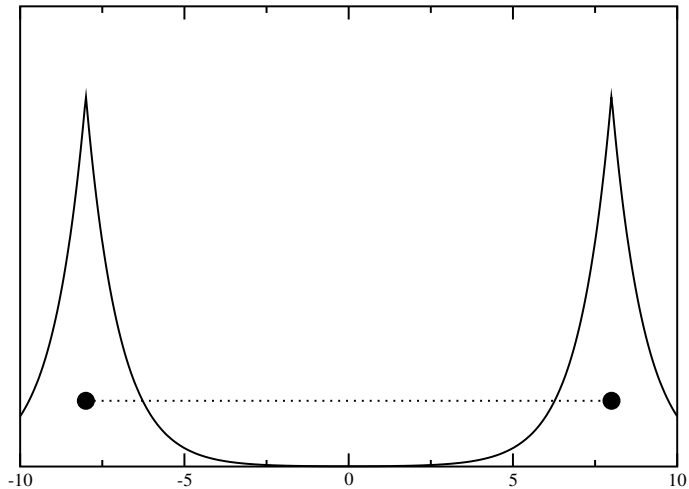
Hydrogen atom: 1s orbital



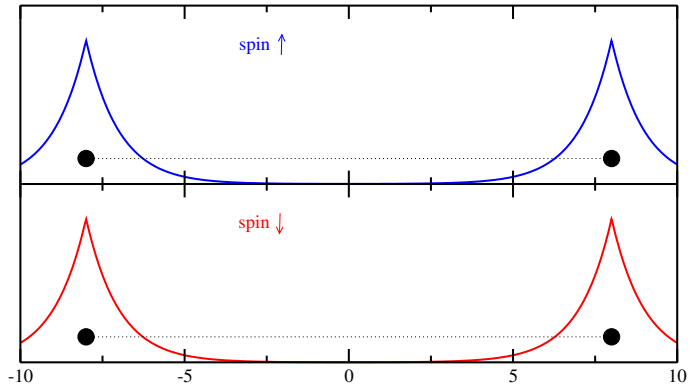
Hydrogen molecule: molecular orbital.



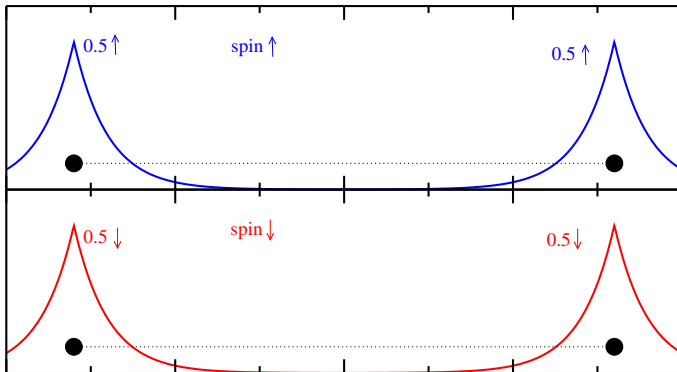
Hydrogen molecule: stretched.



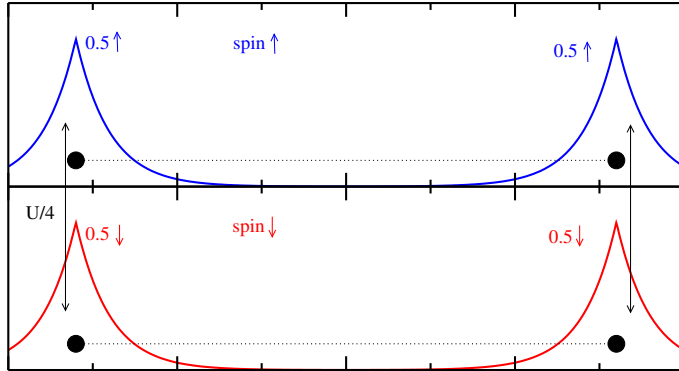
Hydrogen molecule: same orbital for both spins !



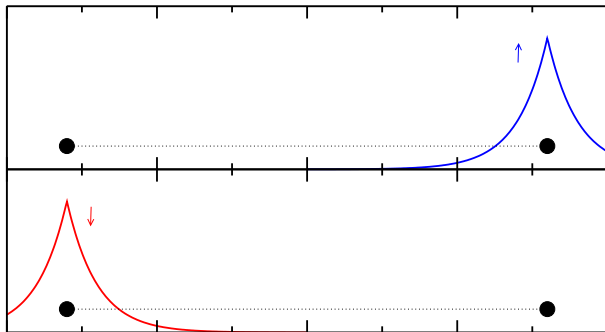
Hydrogen molecule: same orbital for both spins !



Hydrogen molecule: dissociation limit is bad



A solution: break symmetry

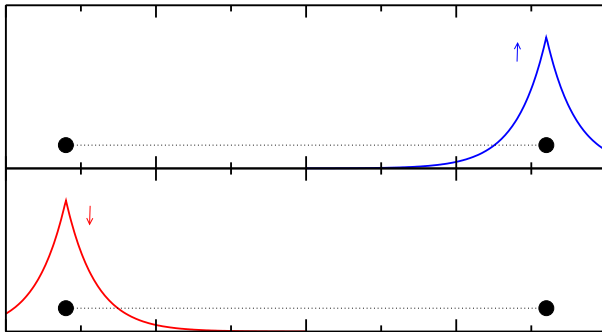


no more artificial delocalization, no more interaction

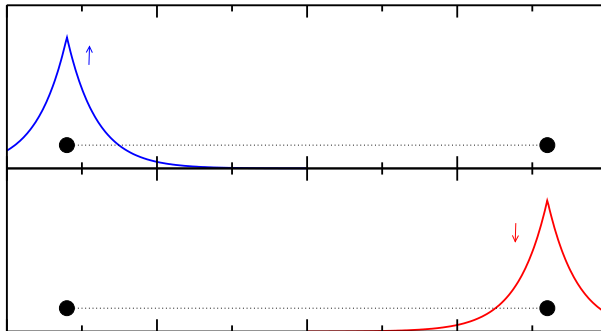
But

- Symmetry breaking. An artificial magnetism is induced.
- A static theory, which overestimates localization.

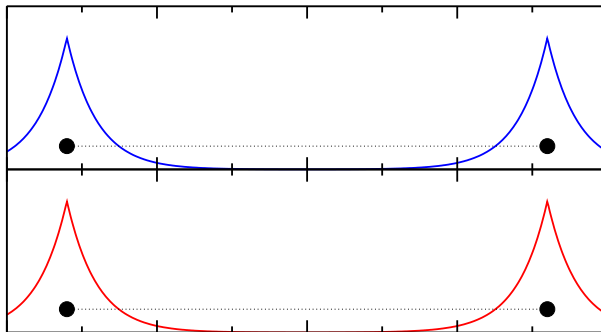
Solution 1: localization: lowers interaction



Solution 2: localization: lowers interaction



Solution 3: delocalization: lowers kinetic energy

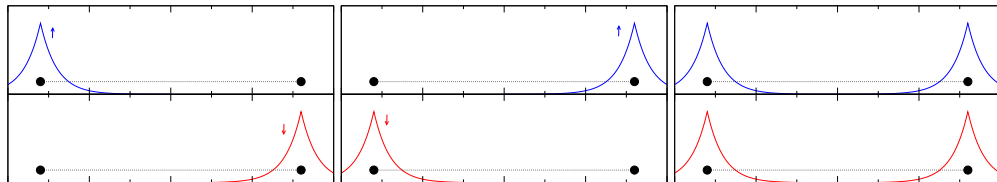


The exact solution

$|\Psi\rangle = c_1|1 : \text{localized } \uparrow\downarrow\rangle$

$+ c_2|2 : \text{localized } \downarrow\uparrow\rangle$

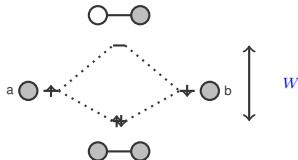
$+ c_3|3 : \text{delocalized}\rangle$



• This mixing of configuration correctly describes the system (magnetism, structural properties).

- $c_1 = c_2$: no ordered magnetism.
- c_3 increases if distance between atoms lowers.

- If $W \gg U$ then $c_1 = c_2 \ll c_3$
- If $W \ll U$ then $c_1 = c_2 \gg c_3$



THE PERIODIC TABLE

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57-71 La-Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89-103 Ac-Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

4s/3d/4p

5s/4d/5p

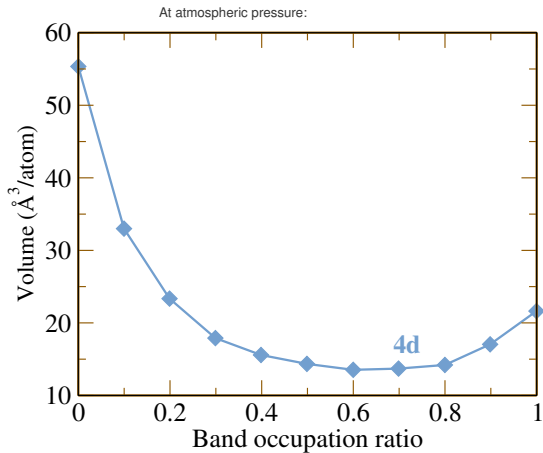
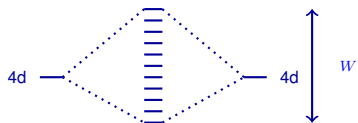
Lanthanides 4f

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
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Actinides 5f

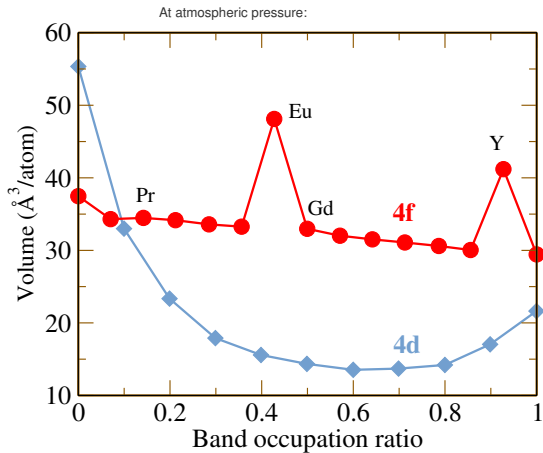
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
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Localization in f electrons systems



[Mac Mahan, et al J. Comp.-Aid. Mater. Des. 5, 131 (1998)]

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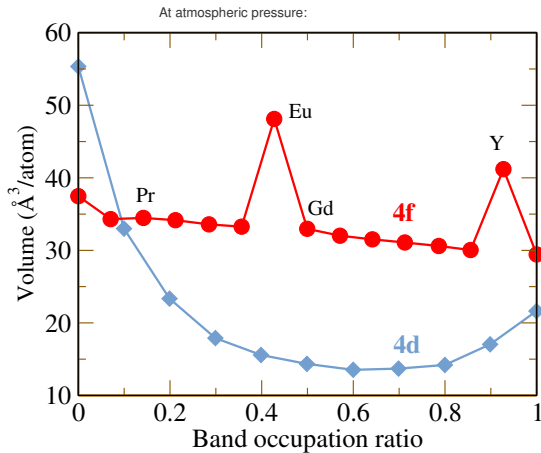
Localization in f electrons systems

4f —·····■·····— 4f $\uparrow\downarrow$ W

Ln $4f^n(6s5d)^3$

Eu $4f^6(6s5d)^3 \Rightarrow 4f^7(6s5d)^2$

Yb $4f^{13}(6s5d)^3 \Rightarrow 4f^{14}(6s5d)^2$



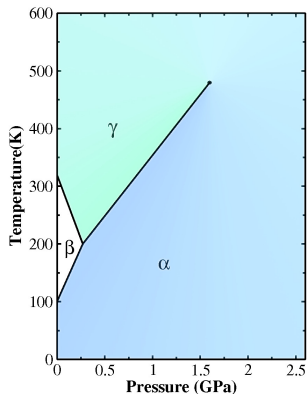
[Mac Mahan, et al J. Comp.-Aid. Mater. Des. 5, 131 (1998)]

Isostructural transition in Cerium

Isostructural transition $\frac{V_\gamma - V_\alpha}{V_\gamma} = 15\%$, ends at a critical point

Electronic configuration $4f^1$.

- α phase: Pauli paramagnetism
 $\Rightarrow \alpha$ phase: f e^- more delocalized.
- γ phase: Curie Paramagnetism
 $\Rightarrow \gamma$ phase: f e^- is localized



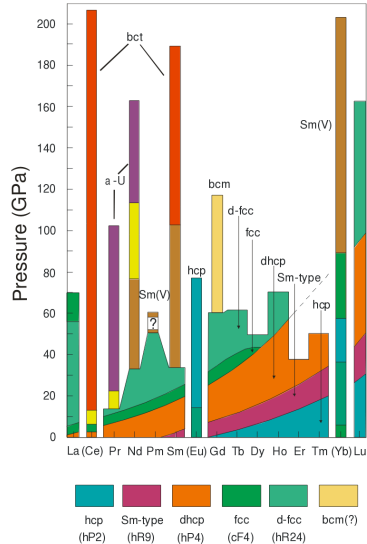
[Johansson, B. Phil. Mag. **30**, 469 (1974)]

Transition in lanthanides.

We now discuss the equation of states of lanthanides as a function of pressure.

- At low pressure, compact structures.
- Under pressure, more distorted structure
 - f electrons participate to the bonding

Figure from [Schiwek, (2002)]

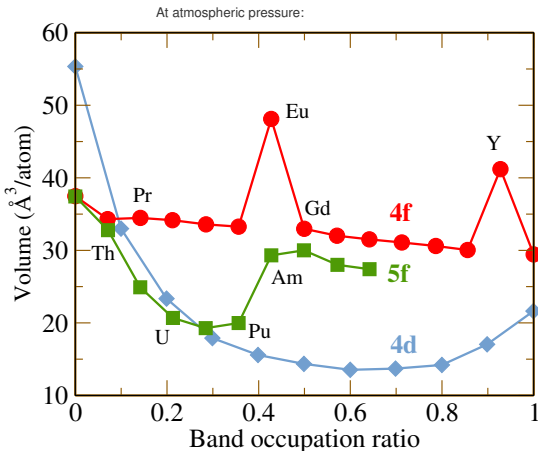


Localization in f electrons systems

4d element: filling of the 4d band (Bonding states and antibonding): 4d electrons are **delocalized**.

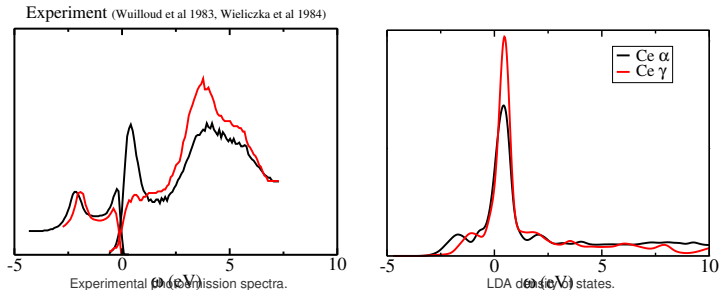
Lanthanides: 4f electrons **localized**, negligible overlap between 4f orbitals.

Actinides: intermediate case of localization.



[Mac Mahan, et al J. Comp.-Aid. Mater. Des. 5, 131 (1998)]

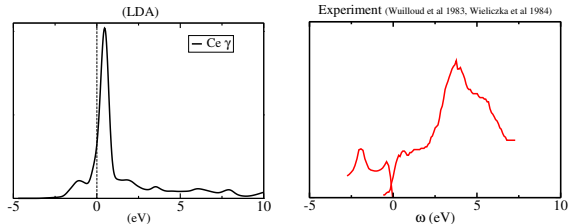
Cerium: experimental spectra and LDA



- Peak at the Fermi level **only in the α phase**.
 γ and α phase: **high energy bands** (-2 eV and 5 eV).
- bands at high energy **not described in LDA**.
- peak at the Fermi level **not correct in LDA**
- $E_{\text{dft-LDA}}(V)$: γ phase **not stable**

γ cerium (paramagnetic)

- Spectral functions: basic features are not reproduced.

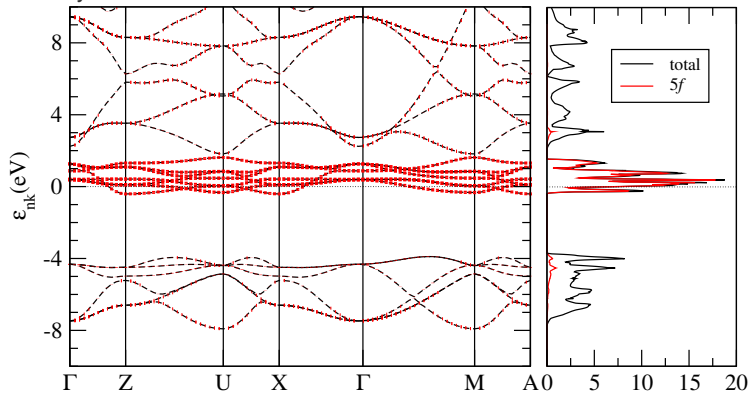


- Structural data

	Exp	DFT
alat (au)	9.76	8.54
B_0 (GPa)	19	55

GGA DOS in UO_2

Density of states in UO_2



THE EXACT HAMILTONIAN

The exact hamiltonien is (i, j are electrons)

$$H = \sum_{i=1}^N \left[-\frac{1}{2} \nabla_{\mathbf{r}_i}^2 + V_{\text{ext}}(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

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Let's simplify this exact hamiltonian

- One orbital per atom: i and j indicates thus only the atom on which orbitals are centered.
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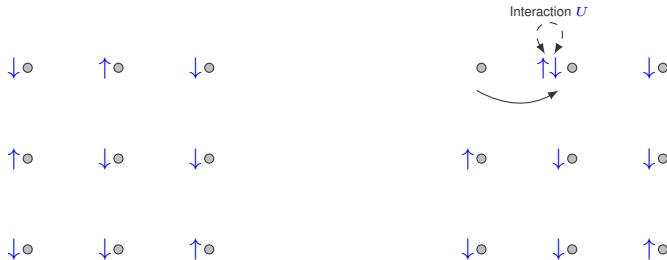
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$$H = \sum_i \epsilon_0 \left(\underbrace{n_{i\uparrow} + n_{i\downarrow}}_{c_{i\uparrow}^\dagger c_{i\uparrow}} \right) + \underbrace{\sum_{j \neq i, \sigma=\uparrow, \downarrow} t_{ij} c_{i\sigma}^\dagger c_{j\sigma}}_{\text{one electron term : delocalization}} + \underbrace{\sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{interaction term : localization}}$$

The Hubbard model: Competition between localization and delocalization

$$H = \underbrace{\sum_{\mathbf{R}, \mathbf{R}'} t_{\mathbf{R}, \mathbf{R}'} c_{\mathbf{R}}^{\dagger} c_{\mathbf{R}'}}_{\text{one electron term : delocalization}} + \underbrace{\sum_{\mathbf{R}} U \hat{n}_{\mathbf{R}\uparrow} \hat{n}_{\mathbf{R}\downarrow}}_{\text{interaction term : localization}}$$



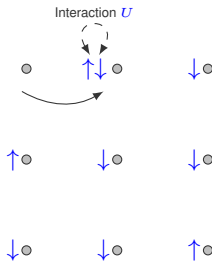
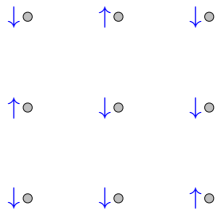
U is the energy repulsion of two electrons on the same site.

For $U/t \gg 1$, the cost of a hopping is $E(|\uparrow\downarrow\rangle) + E(|0\rangle) - 2E(|\uparrow\rangle) = U$

- For large value of the interaction U , electrons are localized
- For low value of the interaction U , electrons are delocalized

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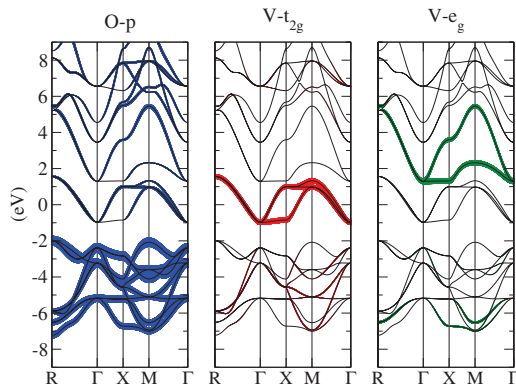
To describe the localized system, a solution is to use the penalty term

$$U n_{\uparrow} n_{\downarrow}$$

Where $n_{\uparrow}$ and $n_{\downarrow}$ are simple numbers of electron on the atom.

The correlated orbital density matrix

- Weight (in red) of Vd orbitals for each wavefunctions. It is $\langle \phi_{n\mathbf{k}} | \chi_{Vd,m} \rangle$.
- How to compute the total number of electron in the Vd orbitals with $m = -2$?

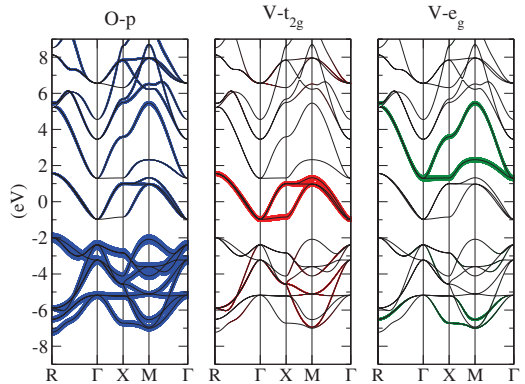


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$$n_{Vd,m=-2} = \sum_{n\mathbf{k}} f_{n\mathbf{k}} |\langle \phi_{n\mathbf{k}} | \chi_{Vd,m=-2} \rangle|^2$$

- Would it be possible to define $n_{Vd,m,m'}$?



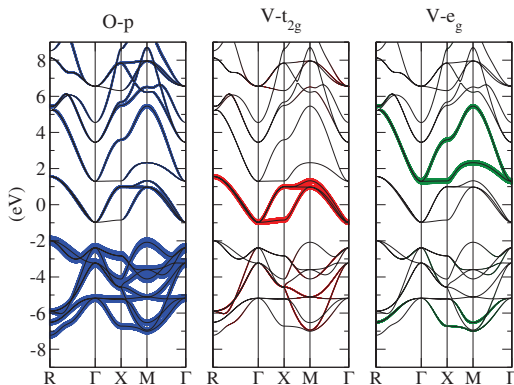
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- Would it be possible to define $n_{Vd,m,m'}$?
Yes, it is just that some KS wavefunctions have contributions from m and m' .

$$n_{Vd,m,m'} = \sum_{n\mathbf{k}} f_{n\mathbf{k}} \langle \phi_{n\mathbf{k}} | \chi_{Vd,m} \rangle \langle \chi_{Vd,m'} | \phi_{n\mathbf{k}} \rangle$$



Rotationnaly invariant DFT+U [Lichtenstein et al (1995)]

The interaction part corresponds to :

$$E_{\text{HF}}^{\text{Coulomb}} = \frac{1}{2} \sum_{i,j} \left[\int \phi_i(\mathbf{r}) \phi_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}) \phi_j(\mathbf{r}') - \int \phi_i(\mathbf{r}) \phi_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \phi_j(\mathbf{r}) \phi_i(\mathbf{r}') \right]$$

We expand the wavefunctions on a new basis: the basis of Vanadium d orbitals and Oxygen p orbitals:

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$$|\phi_i\rangle = \sum_{d,m=-2\dots 2} \langle \chi_{d,m} | \phi_i \rangle | \chi_{d,m} \rangle + \underbrace{\left(\sum_{p,m=-1\dots 1} \langle \chi_{p,m} | \phi_i \rangle | \chi_{p,m} \rangle \right)}_{\text{neglected}} + \dots\dots$$

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and thus

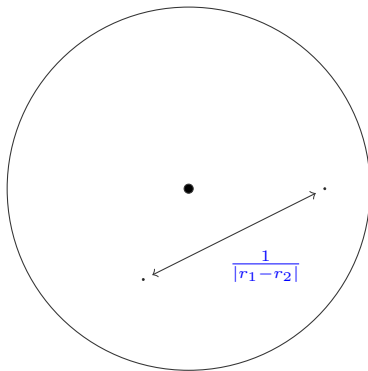
$$E_{\text{HF}}^{\text{Coulomb}} = \frac{1}{2} \sum_{m_1, m_2, m_3, m_4} \left[\langle \chi_{m_1} \chi_{m_2} | \frac{1}{|\mathbf{r} - \mathbf{r}'|} | \chi_{m_3} \chi_{m_4} \rangle n_{m_4, m_2} n_{m_3, m_1} - \langle \chi_{m_1} \chi_{m_2} | \frac{1}{|\mathbf{r} - \mathbf{r}'|} | \chi_{m_3} \chi_{m_4} \rangle n_{m_3, m_2} n_{m_4, m_1} \right]$$

$$n_{Vd, m_1, m_2} = \sum_i \langle \phi_i | \chi_{d, m_1} \rangle \langle \chi_{Vd, m_2} | \phi_i \rangle$$

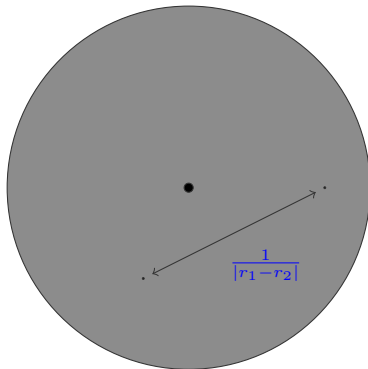
In the cubic symmetry, density matrices are diagonal and thus

$$E_{\text{HF}}^{\text{Coulomb}} = \frac{1}{2} \sum_{m_1, m_2} [\langle \chi_{m_1} \chi_{m_2} | V | \chi_{m_1} \chi_{m_2} \rangle n_{m_2, m_2} n_{m_1, m_1} - \langle \chi_{m_1} \chi_{m_2} | V | \chi_{m_2} \chi_{m_1} \rangle n_{m_2, m_2} n_{m_1, m_1}]$$

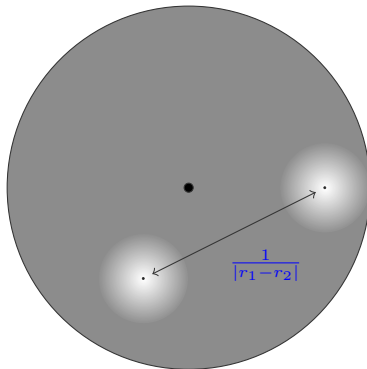
CALCULATION OF U AND SCREENING



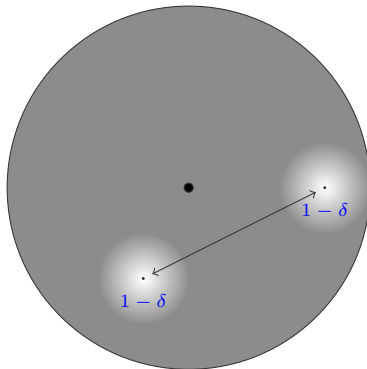
CALCULATION OF U AND SCREENING



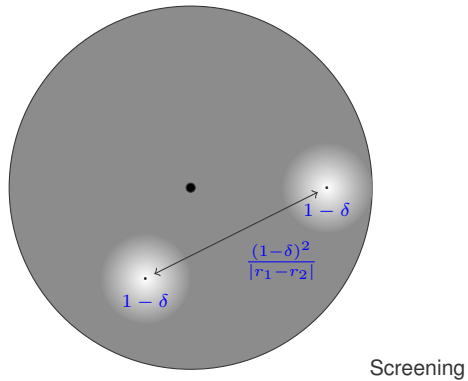
CALCULATION OF U AND SCREENING



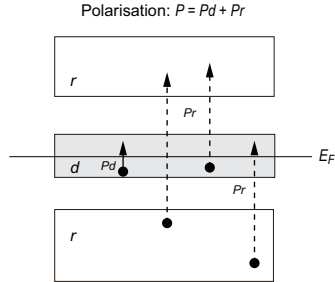
CALCULATION OF U AND SCREENING



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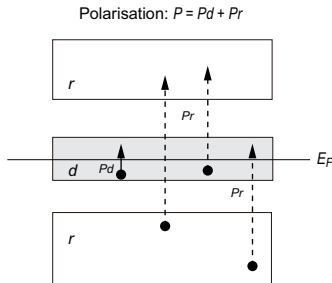


The 1st order perturbation approach (also called cRPA).



- How could we compute the screening ?

The 1st order perturbation approach (also called cRPA).

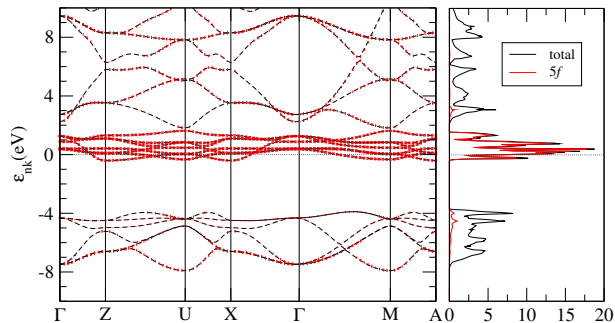


- How could we compute the screening ?
- The interaction is treated as a perturbation and all excitations are taken into account **except the one belonging to the correlated subshell.**

$$\epsilon_r(\omega) = 1 - vP_r(\omega).$$

Effective interactions in UO_2

v (bare)	U
17.1	6.2



Other methods

U is the **screened** interaction between electrons:

- Direct calculation in LDA by **constraint LDA** ¹
 - The coupling between **d** electrons and others is removed for the calculation.
- Direct approach by **linear response theory** ³
 - The **d** local potential is modified.
- cRPA Calculation using the screening from LDA (**GW formalism** ²)
 - Frequency dependent interaction.
- Empirical determination.

(1) Anisimov and Gunnarsson PRB **43** 7570 (1991)

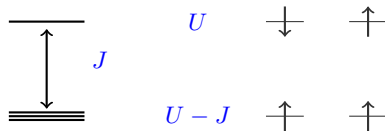
(2) Aryasetiawan, *et al* PRB **70** 195104 (2004)

(3) Cococcioni and de Gironcoli PRB **71** 035105 (2006)

DFT+U

Thus , in the simplest case (restoring spin):

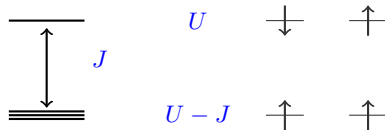
$$E_{\text{HF}}^{\text{U}} = \frac{1}{2} \sum_{m_1, m_2, \sigma} U_{m_1, m_2} n_{m_2}^{\sigma} n_{m_1}^{-\sigma} + \sum_{m_1, m_2, \sigma} (U_{m_1, m_2} - J_{m_1, m_2}) n_{m_2}^{\sigma} n_{m_1}^{\sigma}$$



DFT+U

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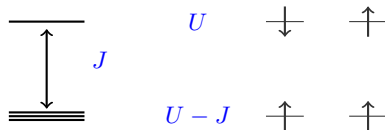
Let's simplify even more, neglecting J and using a constant U .

$$E_{\text{HF}}^{\text{U}} = \frac{1}{2} \sum_{m_1, m_2} U n_{m_2} n_{m_1} = \sum_{m_1 > m_2} U n_{m_2} n_{m_1}$$

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If we have $2l + 1$ orbitals, the number of interactions is $2l(2l + 1)$. The interaction is taken into account twice ! One in E_{xc} , one in E^{U} ! We need to cancel the DFT contribution, a proposition is:

$$E_{\text{HF}}^{\text{dc(=double counting)}} = \frac{N(N - 1)}{2} \text{ with } N = \text{Tr}[n_{m_1}]$$

The DFT+U method

$$E_{\text{DFT}+U} = E_{\text{DFT}} - \overbrace{U \frac{N(N-1)}{2}}^{E_{\text{DC}}} + \overbrace{\frac{U}{2} \sum_{m \neq m'} n_m n_{m'}}^{E_{ee}} = U \sum_m (n_m - n_m^2)$$

In the atom ($n_f=1,0$) $E_{ee} = E_{\text{DC}} = U \frac{N(N-1)}{2}$

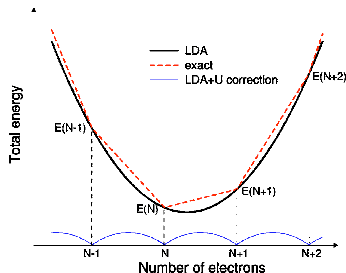


FIG. 1. (Color online) Sketch of the total energy profile as a function of number of electrons in a generic atomic system in contact with a reservoir. The bottom curve is simply the difference between the other two (the LDA energy and the “exact” result for an open system).

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⇒ Self-interaction correction.

From Cococcioni *et al* PRB **71** (2005)

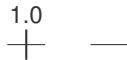
see also Solovyev *et al* PRB **50** 16861 (1994)

Basic effects of the static mean field approximation (DFT+U)

$$E^U = \sum_{m_1 > m_2} U n_{m_2} n_{m_1}$$

- U favors integer occupation of orbitals
- It penalizes non integer values.

$$E_U = 0 \times 1U = 0$$



$$E^U = 0.5 \times 0.5U = 0.125U$$



Basic effects of the static mean field approximation (DFT+U)

$$E = E_{\text{DFT}} + \frac{U}{2} \sum_{m \neq m'} n_m n_{m'} - U \frac{N(N-1)}{2}$$

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$$\frac{\partial n_m}{\partial \phi_{n\mathbf{k}}^*} = \frac{\partial \left[\sum_{n\mathbf{k}} \langle \phi_{n\mathbf{k}} | \chi_m \rangle \langle \chi_m | \phi_{n\mathbf{k}} \rangle \right]}{\partial \phi_{n\mathbf{k}}^*} = |\chi_m\rangle \langle \chi_m | \phi_{n\mathbf{k}}\rangle$$

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So gathering the terms:

$$\frac{\partial E}{\partial \phi_{n\mathbf{k}}^*} = \sum_m = |\chi_m\rangle U \left(\frac{1}{2} - n_m \right) \langle \chi_m | \phi_{n\mathbf{k}}\rangle$$

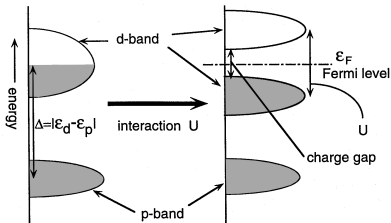
Basic effects of the static mean field approximation (DFT+U)

$$E = E_{\text{DFT}} - U \frac{N(N-1)}{2} + \frac{U}{2} \sum_{m \neq m'} n_m n_{m'} \Rightarrow V = V_{\text{DFT}} - \sum_m |\chi_m\rangle U (n_m - \frac{1}{2}) \langle \chi_m|$$

- $n_m = 0$, $V = V_{\text{DFT}} + \frac{U}{2}$
- $n_m = 1$, $V = V_{\text{DFT}} - \frac{U}{2}$

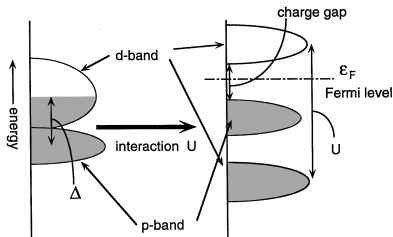
⇒ A gap is opened among correlated orbitals.

Charge transfert insulators and Mott Hubbard insulators



(a) Mott-Hubbard Insulator

Mott insulators: Gap excitations are d-d (or f-f) ie between **Hubbard bands**.



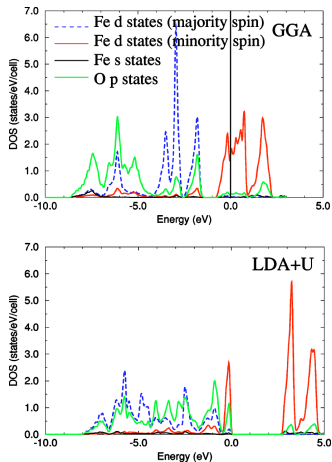
(b) Charge Transfer Insulator

Charge Transfert insulators: Gap excitations are **Op-d (Or Op-f)**
Correlation opens the gap in the two cases !

Imada RMP (1998)

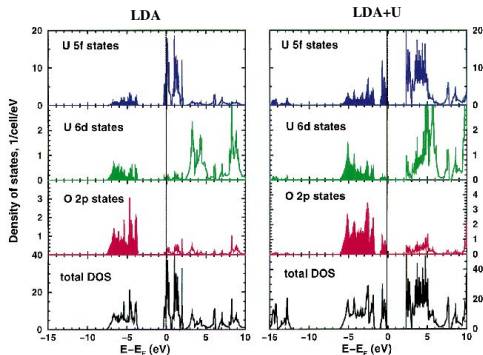
Zaanen, Sawatzky, and Allen PRL (1985)

Main effects of Mean Field approximation (DFT+U)



FeO (d^6): insulator in DFT+U

Cococcioni et al PRB 71 2005



UO₂ (f^2): antiferromagnetic, insulator

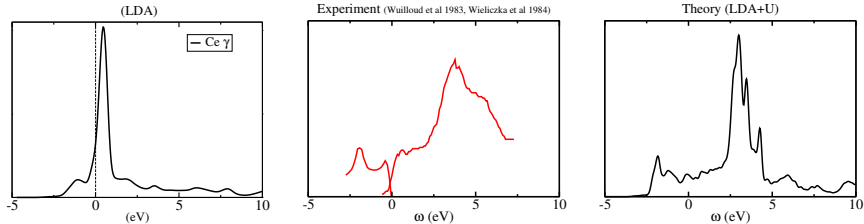
Gap_{exp}=2.1 eV

electrons localization: volume increases in DFT+U

Dudarev et al Micron 31 2000

γ cerium (paramagnetic)

- Spectral functions: basic features are reproduced.



- Structural data
⇒ Electron repulsion induces a weakening of the bonding.

(1) Shick, Pickett, Lichtenstein 2000,
Amadon, Jollet, Torrent PRB 2008.

	Exp	DFT+U ¹	DFT
alat (au)	9.76	9.83/9.54	8.54
B_0 (GPa)	19	29.6/34	55

- But: The α phase is not correctly described, magnetism is incorrect (except for the β phase), no transitions.

Calculation of $\langle m_1 m_3 | V | m_2 m_4 \rangle$

$$E_{\text{HF}}^{\text{Coulomb}} = \frac{1}{2} \sum_m \left[\langle \chi_{m_1} \chi_{m_2} | \frac{1}{|\mathbf{r} - \mathbf{r}'|} | \chi_{m_3} \chi_{m_4} \rangle n_{m_4, m_2} n_{m_3, m_1} - \langle \chi_{m_1} \chi_{m_2} | \frac{1}{|\mathbf{r} - \mathbf{r}'|} | \chi_{m_3} \chi_{m_4} \rangle n_{m_3, m_2} n_{m_4, m_1} \right]$$

One uses $\frac{1}{r_{12}} = \sum_{k=0}^{\infty} \sum_{m=-k}^k \frac{4\pi}{2k+1} \frac{r_{<}^k}{r_{>}^{k+1}} Y_k^m(\theta_1, \phi_1) Y_k^{m*}(\theta_2, \phi_2)$ and after some manipulations:
 $\langle m_1 m_3 | V_{\text{ee}} | m_2 m_4 \rangle$ contains an angular and a radial part.

$$\langle m_1 m_3 | V_{\text{ee}} | m_2 m_4 \rangle = 4\pi \sum_{k=0,2,4,6} \frac{F_k}{2k+1} \sum_{m=-k}^{+k} \langle m_1 | m | m_2 \rangle \langle m_3 | m | m_4 \rangle$$

$$U = \frac{1}{(2l+1)^2} \sum_{m_1, m_2} \langle m_1 m_2 | V_{\text{ee}} | m_1 m_2 \rangle = F_0 \text{ coulomb term}$$

$$J = \frac{1}{2l(2l+1)} \sum_{m_1 \neq m_2} \langle m_1 m_2 | V_{\text{ee}} | m_2 m_1 \rangle = \frac{F_2 + F_4}{14} \text{ exchange term}$$

The DFT+U method

$$E_{\text{DFT+U}} = E_{\text{DFT}} - \overbrace{U \frac{N(N-1)}{2}}^{E_{\text{DC}}} + \overbrace{\frac{U}{2} \sum_{m \neq m'} n_m n_{m'}}^{E_{ee}} = U \sum_m (n_m - n_m^2)$$

In the atom ($n_f=1,0$) $E_{ee} = E_{\text{DC}} = U \frac{N(N-1)}{2}$

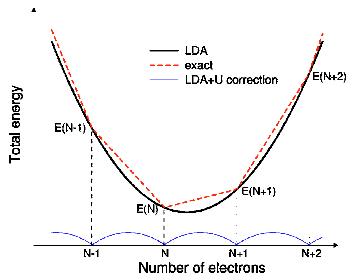


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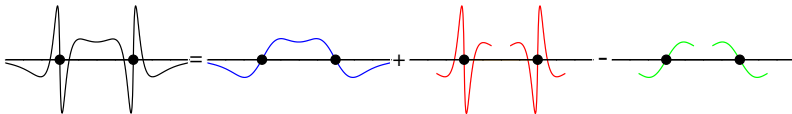
From Cococcioni *et al* PRB **71** (2005)

see also Solov'yev *et al* PRB **50** 16861 (1994)

The PAW method: Expression of a Kohn Sham function of Cl_2 .

$$|\Psi_{\mathbf{k}\nu}\rangle = \underbrace{|\tilde{\Psi}_{\mathbf{k}\nu}\rangle}_{\text{On plane waves}} + \underbrace{\sum_i |\varphi_i\rangle \langle \tilde{p}_i | \tilde{\Psi}_{\mathbf{k}\nu}\rangle - \sum_i |\tilde{\varphi}_i\rangle \langle \tilde{p}_i | \tilde{\Psi}_{\mathbf{k}\nu}\rangle}_{\text{On a local radial grid}}$$

Blöchl PRB 1994



[Developed in ABINIT by the CEA group]

[M. Torrent, F. Jollet, F. Bottin, G. Z  rah, X. Gonze Comp. Mat. Science 42 (2), 337-351 (2008)]

A grid devoted to local properties : well adapted to correlated systems and to compute:

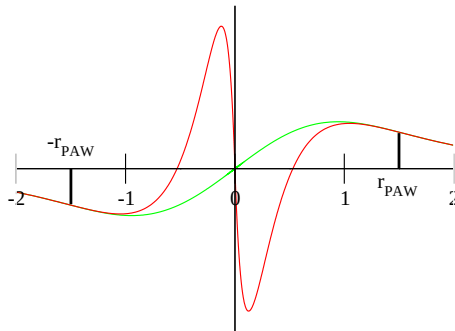
- DFT+ U density matrix.
- Projected Wannier orbitals.

DFT: The PAW method

A Kohn-Sham function can be written:

$$|\Psi_{\mathbf{k}\nu}\rangle = \underbrace{|\tilde{\Psi}_{\mathbf{k}\nu}\rangle}_{\text{On plane waves}} + \underbrace{\sum_i |\varphi_i\rangle \langle \tilde{p}_i | \tilde{\Psi}_{\mathbf{k}\nu}\rangle - \sum_i |\tilde{\varphi}_i\rangle \langle \tilde{p}_i | \tilde{\Psi}_{\mathbf{k}\nu}\rangle}_{\text{On a local radial grid}}$$

- φ_i : atomic wavefunction
- $\tilde{\varphi}_i$: pseudo atomic wavefunction



ABINIT keywords for DFT+*U*

```
# == LDA+U
usepawu 1          # activate DFT+U
lpawu 2 -1         # apply DFT+U for d orbitals for the first species only
upawu 8.0 0.0      # Value of U
jpawu 1.0 0.0      # Value of J

usedmatpu 10       # Number of steps to impose a density matrix
dmatpawu
  1.0 0.0 0.0 0.0 0.0 # up density matrix
  0.0 1.0 0.0 0.0 0.0 # up density matrix
  0.0 0.0 1.0 0.0 0.0 # up density matrix
  0.0 0.0 0.0 1.0 0.0 # up density matrix
  0.0 0.0 0.0 0.0 1.0 # up density matrix

  1.0 0.0 0.0 0.0 0.0 # dn density matrix
  0.0 1.0 0.0 0.0 0.0 # dn density matrix
  0.0 0.0 0.0 0.0 0.0 # dn density matrix
  0.0 0.0 0.0 1.0 0.0 # dn density matrix
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```

Conclusion

- Strong correlations
 - Localized orbitals induce strong electronic Coulomb interaction.
 - This repulsion can induce strong localization of electrons and creates Mott insulator.
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 - Hubbard bands are signatures of strong interaction.
- DFT+ U
 - Describe well magnetically ordered phases.
 - Magnetic Mott insulators.
- **Static** Mean Field solution : Fixed (frozen) occupancies.
 - Magnetic order: Paramagnetic insulators cannot be described.
 - V_2O_3 , γ Cerium.
 - Orbital order and anisotropy are overestimated
 - Electrons are frozen.
 - Metallic and correlated phase are out of reach (α Cerium, $SrVO_3$).
 - Transition induced by localization cannot be described.

Spectral functions of cerium

Theoretical spectral functions compared to photoemission spectra

